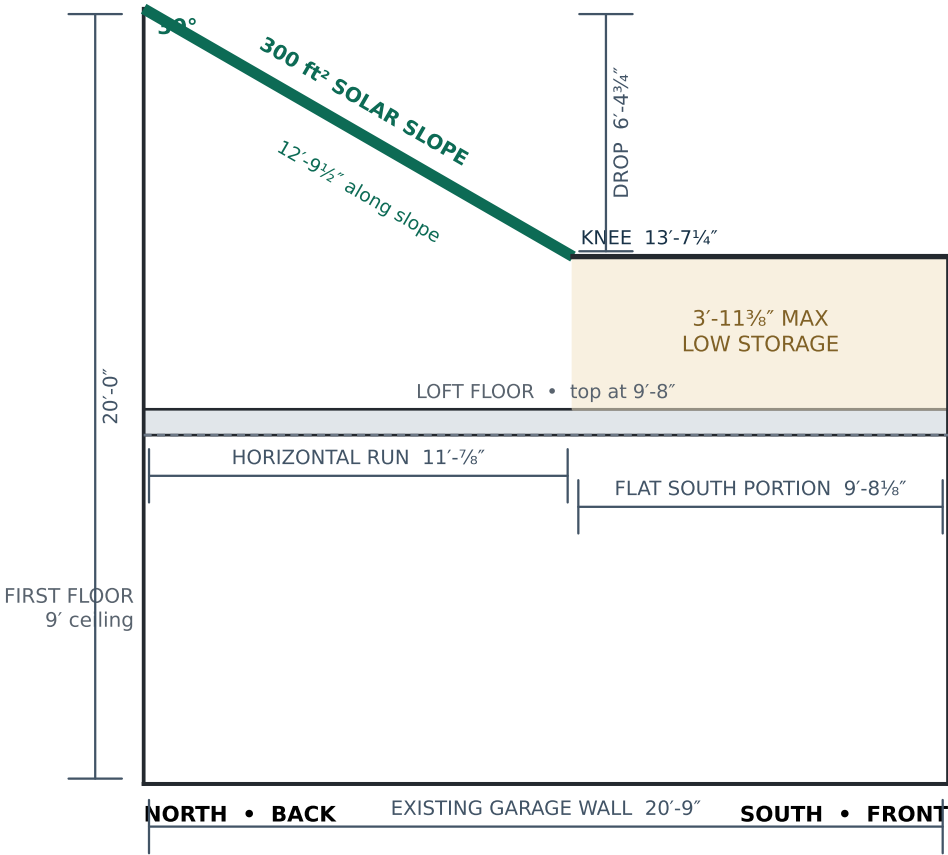


WEST ELEVATION — 300 ft² SOLAR-ROOF ENVELOPE STUDIES

Assumes the solar surface spans the full 23'-5½" east-west width • elevations measured from grade

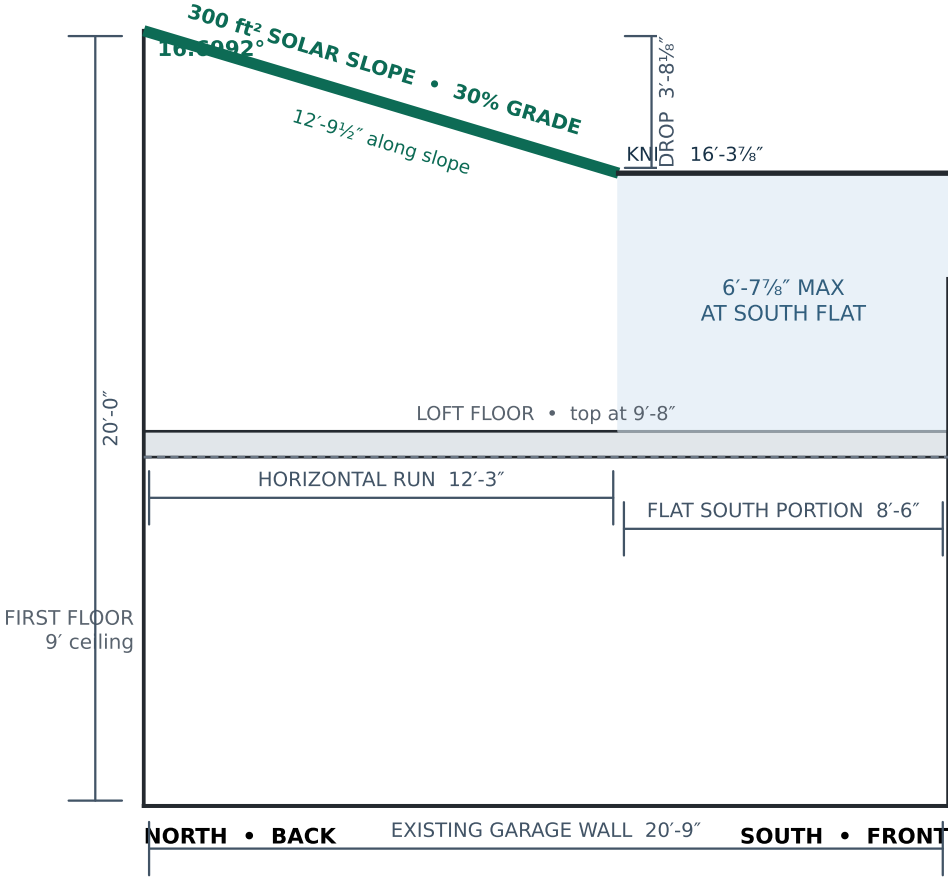
A — 30° slope beginning at the north wall

Baseline requested envelope; southern loft becomes low storage space



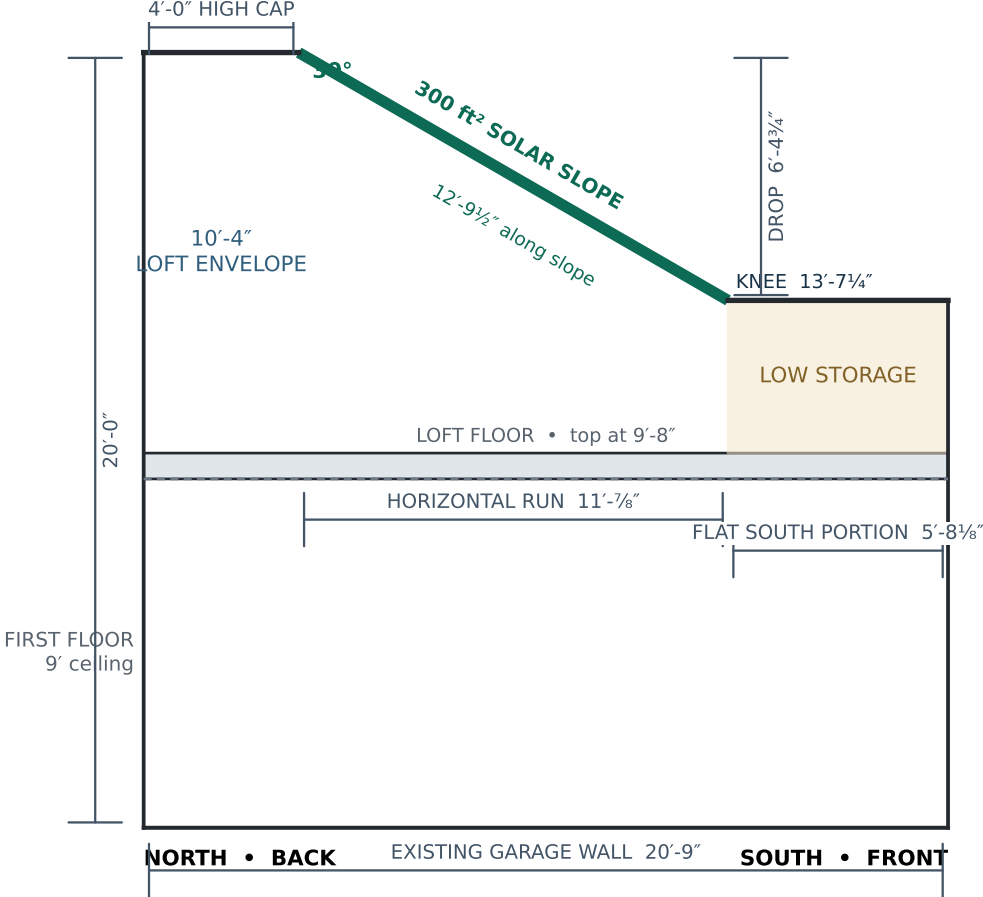
B — If “30%” is literal

A 30% grade is 16.7°, not 30°; this leaves much more useful front height



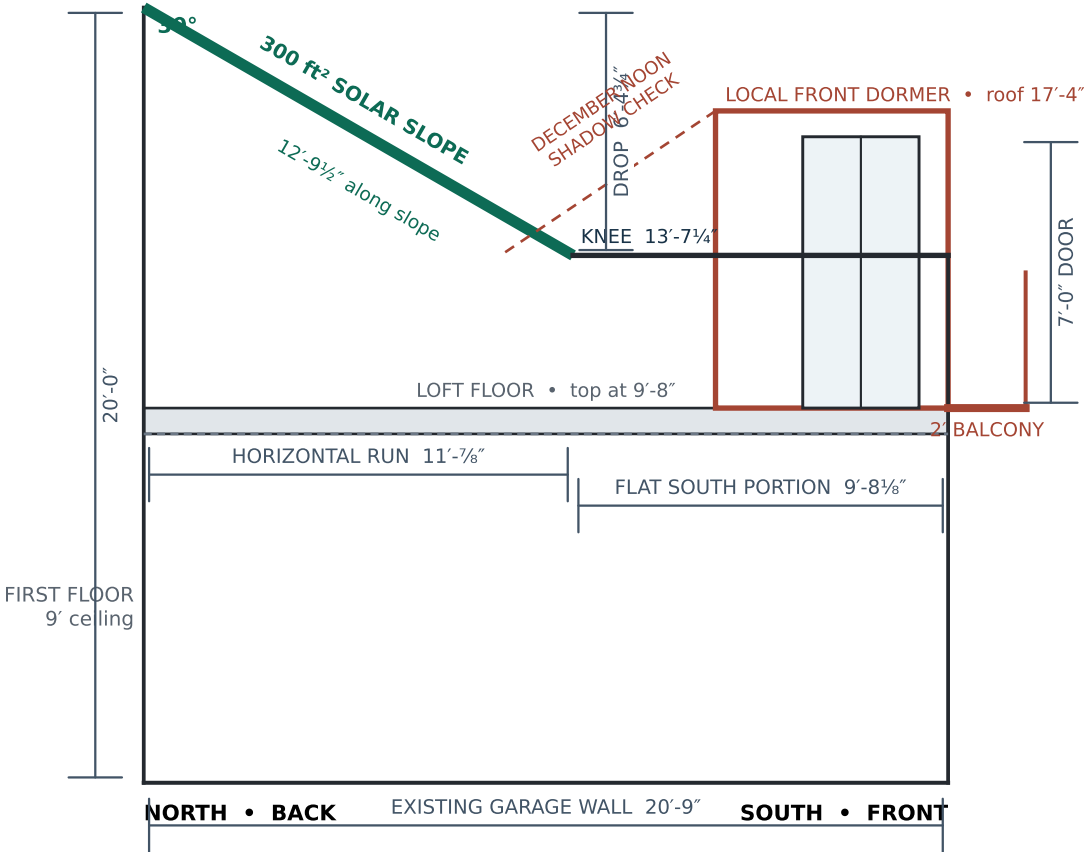
C — Move the 30° solar band 4 ft south

Adds a full-height north cap while preserving the same 300 ft² solar plane



D — 30° slope with a usable south balcony dormer

A loft-level balcony needs a local raised front volume; dashed line shows winter-noon shadow



Concept envelope only. Roof thickness, drainage pitch, parapets, structure and required clearances are not yet included. A dormer or balcony occupying part of the solar plane reduces generating area; option D places the raised volume south of the 300 ft² band but introduces seasonal shadow.